QELO-X | CORE ORCHESTRATION

SECURE ENGINE v1.1.0

ARCH_TYPE: DISTRIBUTED_NODE_PROXY | ENCRYPTION: AES-256-GCM | OS: LINUX_KERNEL_HARDENED

This document details the advanced engineering patterns of **QELO-X**, the security-first orchestrator for the Quai Network. Designed for high-authority node operators, the system prioritizes **DevSecOps** principles: Continuous Monitoring, Automated Recovery, and Hardened Networking.

FIGURE 1.0: HOLOGRAPHIC INTERFACE FOR REAL-TIME NODE TELEMETRY

>_Network & Architecture Topology

The architecture is built on a non-blocking Go runtime, orchestrating low-level C++ binaries (go-quai) while providing a high-level secure management layer.

```
graph LR A[External Traffic] -->|TCP 9000| B(Hardware Firewall) B -->|Auth Required| C{QELOX Proxy} C -->|Unix Socket| D(Quai Engine) C -->|HTTPS/TLS| E[Web Analytics] D -->|P2P| F((Global Mesh)) style C fill:#003366,stroke:#00f3ff,stroke-width:2px style D fill:#004400,stroke:#00ff41,stroke-width:2px
```

THREAD_POOL_STATUS

OPTIMIZED [Go-Routines]

SOCKET_PERMISSION_MASK

0660 (USER+GROUP ONLY)

>_Security & Hardening (DevSecOps)

Security is not a feature; it's the foundation. QELO-X implements several layers of defense-in-depth:

[√] **Isolated Config Tracking:** Sensitive keys and RPC secrets are strictly excluded from SCM through advanced Git filtering.
[√] **Subprocess Jail:** Node processes are strictly managed via process IDs, preventing "Zombie" processes and orphan leaks.
[√] **Health Score Heuristics:** Real-time mathematical scoring (0-100) based on peer stability, block latency, and system IOPS.

FIGURE 2.0: NEURAL-NETWORK BASED INFRASTRUCTURE MONITORING

>_Automated Deployment Pipeline

sequenceDiagram participant Dev as DevOps Engineer participant Git as Repository (GitHub) participant QX as QELO-X Daemon participant S as Storage Dev->>Git: Push Secure Update Git->>QX: Webhook Trigger QX->>S: Backup Config.toml QX->>QX: Internal Health Check (Pre-flight) QX->>QX: Safe Restart sequence QX-->>Dev: Alert: Deploy Success

FIGURE 3.0: INTEGRATED TERMINAL ENVIRONMENT FOR ADVANCED SSH OPERATORS